

# Frameworks Linking Six Core Primitives to Emergence

In recent decades, several frameworks across biology, computation, and systems theory have identified fundamental “primitives” or mechanisms that echo the six themes given – **(P1)** operator rewriting, **(P2)** constraint gating, **(P3)** protocol/phase dependence, **(P4)** invariant classes (sectors), **(P5)** closure/encapsulation, and **(P6)** resource-driven asymmetry/affinity. Below we survey prominent examples, explaining how each framework embodies (or can be mapped to) these six ingredients, and whether it explicitly addresses *emergence*, open-ended evolution, or irreversibility. We note whether the mapping of P1–P6 is **strong** (explicit in the framework) or **speculative/analogical** (an interpretive overlay). All cited sources are post-1950s, as requested.

## Complex Adaptive Systems (CAS) – Holland’s “Building Blocks” (1990s)

John H. Holland’s work on Complex Adaptive Systems provides a well-known framework that, while not explicitly enumerating *six* primitives, identifies several fundamental mechanisms overlapping with P1–P6. Holland described **seven “basics”** of CAS (aggregation, tagging, nonlinearity, flows, diversity, internal models, building blocks) <sup>1</sup> <sup>2</sup>. Many of these map onto the primitives:

- **Aggregation & Building Blocks:** CAS aggregate lower-level components into higher-level *building blocks* <sup>3</sup> <sup>2</sup>. This is analogous to **closure/packaging (P5)** – combining agents or schema into encapsulated units that can act as single entities. It provides a form of modularity and “closure” in that the aggregate behaves as a unified whole.
- **Tagging (Selective Interaction):** *Tags* are markers that mediate and **gate interactions** among agents <sup>4</sup>. Tags “manipulate symmetries” by making certain interactions possible while filtering out others <sup>4</sup>. This corresponds directly to **constraint gating (P2)** – only agents with matching tags interact, imposing context-dependent constraints on who can affect whom. Tagging is explicitly cited as a mechanism for hierarchical organization (higher-order agents form via tags) <sup>5</sup>, facilitating emergent structures.
- **Nonlinearity & Flows:** CAS behaviors are **order-dependent and nonlinear**, meaning the sequence of interactions (and feedback loops) matters for the outcome <sup>6</sup>. Holland notes that “behavior in aggregate is more complex than the parts” <sup>7</sup>, underscoring **protocol/phase dependence (P3)** – the path of interactions (protocol) can lead to different emergent states. *Flows* of resources through networks of agents <sup>8</sup> embody **resource-driven asymmetry (P6)**: injected resources propagate and cause cascades (the “multiplier effect”) <sup>9</sup>, and recycling of resources creates feedback that drives open-ended growth of diversity <sup>10</sup>. These flows ensure CAS are *far from equilibrium*, a prerequisite for sustained emergence.

- **Internal Models & Diversity:** CAS agents have internal models and adapt via **variation and selection**, maintaining **invariant patterns (P4)** at higher levels even as components adapt <sup>11</sup> <sup>2</sup> . For example, a stable niche or an evolved “tagged” group can be seen as an invariant class of agent strategies that persists while details change. Diversity of agent types provides multiple such classes and ensures novelty (no single state absorbs the system) <sup>12</sup> .

Overall, CAS theory **strongly addresses emergence**: simple local rules produce complex, coherent macro-behaviors <sup>4</sup> . Holland explicitly frames CAS as explaining “coherence under change” – how unstructured agents self-organize <sup>13</sup> . The mapping of CAS mechanisms to P1–P6 is **moderate**: P2, P5, and P6 are very direct (tagging, aggregation, resource flows), while P3 and P4 appear as properties of the dynamics (path-dependent interactions and persistent patterns). **Operator rewriting (P1)** is less explicit in CAS; however, one could view genetic algorithm operators or recombination of building blocks as a kind of rule-based generation of new solutions – a **speculative** mapping. In summary, CAS frameworks like Holland’s provide a solid example of multiple primitives coordinating to yield emergent complexity <sup>1</sup> <sup>8</sup> .

**Emergence focus:** High. CAS are explicitly studied to understand emergence of macroscopic order (e.g. ant colony intelligence from simple ants) <sup>14</sup> . However, CAS models *in themselves* do not always guarantee **open-ended** evolution (they can reach equilibrium of agent types), but they set the stage for it by emphasizing continual adaptation and novelty introduction <sup>12</sup> .

## Autopoiesis – Maturana & Varela’s Self-Producing Systems (1973)

The concept of **autopoiesis** (Greek for “self-making”) was introduced by Maturana and Varela as a definition of life’s organization <sup>15</sup> <sup>16</sup> . An autopoietic system is *explicitly* defined in terms of a set of primitive processes that map well to P1–P6:

- **Recursive Production (P1 & P5):** An autopoietic system is a *network of component-producing processes* such that the components recursively **generate and regenerate the network that produces them** <sup>15</sup> <sup>16</sup> . In other words, the system’s operations continually *rewrite* or transform raw materials into the very components that sustain the network (analogous to **operator rewriting (P1)** in a chemical or mechanical sense). The system is **operationally closed** – the product of its processes is the maintenance of the process network itself <sup>16</sup> . This is **closure (P5)** in the strongest sense: the system is organizationally closed and produces its own boundary. Maturana’s definition explicitly “operates with a notion of recursive closure” <sup>17</sup> .
- **Boundary and Gating (P2):** The components produced by an autopoietic network include those that *constitute the system’s boundary* <sup>16</sup> . For example, a cell produces the components of its membrane. This boundary both demarcates the system and **regulates interactions** with the environment (only certain molecules pass, etc.). Thus **constraint gating** is inherent – the boundary gates what can enter or leave, and internal process constraints determine which external perturbations have an effect. The system’s **organization remains invariant (P4)** in that any influence from outside can only trigger changes allowed by the internal network’s structure (Maturana & Varela call this *structural determinism*). The *invariant class* here is the organizational pattern (the network topology) that persists even as material components turn over.
- **Protocol/Phase Dependence (P3):** Autopoiesis involves **cyclic processes** – e.g. production of metabolites, assembly of membrane, etc., in a *sequence* that continuously loops. The order of

operations is critical: if the process sequence is broken (e.g. membrane not repaired before it disintegrates, or template not copied at the right time), the system will collapse. In this sense autopoietic systems have an implicit “protocol” (a self-maintenance cycle) that must be followed. This is analogous to a **phase-dependent** process or a “protocol” the system executes to remain alive. (Maturana and Varela didn’t use the word “protocol,” but the idea of a required organizational *sequence* is present in their biological examples, and later commentators describe autopoiesis as involving feedback loops and cyclic phases of operation <sup>15</sup>.)

- **Resource-Driven Asymmetry (P6):** Although autopoietic systems are operationally closed, they are thermodynamically open – they require a **throughput of energy or matter** to sustain their ordered state. A living cell, for instance, must import low-entropy nutrients and export high-entropy waste <sup>18 19</sup>. This creates a **dissipative, far-from-equilibrium** condition (breaking time-reversal symmetry) that drives the self-production processes. Maturana and Varela noted that while the *organization* is independent of specific inputs, **if the inputs (resources) cease, the system decays**. Thus an autopoietic system only exists in conjunction with a suitable energy/matter gradient – a clear case of **resource-driven asymmetry (P6)** powering an irreversible maintenance process.

Overall, autopoiesis hits **P5 and P2 very strongly** (closure and boundary gating are literally in the definition <sup>16</sup>), and maps **P6** clearly (dissipative self-maintenance). P1 is implicit in the transformative production operations. P3 and P4 appear as logical consequences (cyclic organization, invariant pattern under component replacement). The mapping is **strong**, since the framework was formulated in systems terms closely aligned with these primitives.

Importantly, autopoiesis was proposed *to explain the emergence of living units*. It characterizes how **emergence of a self-sustaining individual** is possible – e.g. how chemical reactions could spontaneously form a cell-like unit with a boundary. This is **emergence through closure**: the whole (the autonomous unit) is an emergent entity produced by the network of processes. Autopoiesis theory itself focuses on *sustaining* the living state once formed, rather than open-ended innovation. But it implies irreversibility: assembling an autopoietic network is a path-dependent event (you must go through the right sequence to “boot up” life). If achieved, the system then continuously reproduces itself – a kind of dynamic steady state that resists perturbations (except those that break the closure). In summary, autopoiesis provides a clear set of generative mechanisms for emergence of living organization, with a **strong mapping** to P1–P6 (especially P5, P2, P6) and an emphasis on *organizational emergence*. Sources explicitly note its recursive closure and self-bounded nature <sup>16</sup>.

## Rosen’s Relational Biology – Metabolism-Repair Systems (1958–1991)

The theoretical biologist Robert Rosen developed a formal framework for what he considered the essence of life, using category theory and relational models. His best-known construct is the **(M,R) system**, an abstract metabolism–repair network. Rosen’s framework identifies primitives akin to **closure (P5)**, **invariants (P4)**, and **constraint dependencies** in living systems:

- **Closure to Causation (P5):** Rosen argued that a living organism is characterized by a closed loop of efficient causation. In an (M,R) model, “Metabolism” (M) is a function that produces the components of the system (analogous to the organism’s biochemical network), and “Repair” or replication (R) is a

function that produces or replaces the *metabolic function* itself (e.g. producing the enzymes that carry out metabolism). These two functions are arranged such that **every component of the system is produced by the system itself** – a closure of production and repair. This is directly **closure (P5)**: the system is organizationally closed under its set of transformations. It's essentially the same concept as autopoietic closure, formalized in category-theoretic terms <sup>20</sup>. Every role (function) needed for the system's operation is fulfilled by some internal component, not by an external agent.

- **Invariant Classes (P4)**: In Rosen's relational view, one focuses on the *roles* or *functions* rather than the matter. For instance, "enzyme that catalyzes reaction X" is a functional role that could be filled by different molecules. These roles are **invariant classes** – categories that remain even as specific molecules turn over. The (M,R) system's organization can be seen as an invariant schema of relationships: as long as something plays the part of M and something plays R, the system persists. This resembles **P4 (invariant sectors)**: a class of components (like "all possible catalysts of reaction X") is an invariant set under the system's dynamics. Rosen's approach was to show life is defined by such invariant functional relationships, rather than any particular substance.
- **Constraint Gating and Dependency (P2 & P3)**: The (M,R) system highlights **dependent relationships** – metabolism depends on catalysts (produced by R), and R in turn depends on metabolism's outputs. Thus there is an in-built **protocol/order**: e.g. the catalysts must exist for metabolism to work, and metabolism must produce the building blocks for the repair function. This implies a **phase dependence (P3)** or at least mutual dependency cycle – if the sequence is broken (no catalysts, metabolism stops; no metabolism, catalysts can't be produced), the system fails. In practice, an organism executes a *protocol* of gene expression and metabolic activity to regenerate itself each generation. Additionally, each catalytic reaction in M can be seen as **gated by constraints (P2)** – i.e. it only proceeds if the specific enzyme is present. Rosen's model is abstract, but we can interpret it such that the *presence of the R-product (enzyme)* **gates** the metabolic reaction. There is no *explicit* mention of logic gates, but the formalism's structure has each function only "firing" when its domain input is available, analogous to a constraint that must be satisfied (for example, no catalysis without catalyst).
- **Resource-Driven Asymmetry (P6)**: While Rosen's mathematical models were timeless (he didn't model kinetics directly), he clearly understood that any real metabolism requires a constant throughput of matter and energy from an environment. A metabolism is **dissipative** – it must export entropy to maintain order <sup>18</sup> <sup>19</sup>. Rosen discussed organisms as open systems that *create* organization by processing resources. Thus the idea of an external "food" or resource input driving the system was implicit. The asymmetry (irreversibility) of consumption and waste production underlies why such systems can exist away from equilibrium. So P6 is present as a necessary condition for the M,R schema to correspond to life, though it was not the focus of Rosen's formalism.

Rosen's relational biology strongly emphasizes P5 (closure) and P4 (abstract invariants of organization). P2–P3 are present as relational dependencies (the structure *is* a set of constraints linking functions). The mapping is **moderate**: Rosen himself did not list "six primitives," but his core idea of closure to efficient cause <sup>20</sup> aligns with P5, and he explicitly connected his ideas to others like autocatalytic sets and autopoiesis <sup>20</sup>. In fact, modern researchers note that Rosen's (M,R) systems, autopoiesis, and autocatalytic (RAF) sets are closely related notions of organizational closure <sup>20</sup>.

**Emergence:** Rosen's framework aimed to **explain the emergence of life and its irreducibility**. He argued that a closed causal network (which is non-computable in a certain sense) is what distinguishes life from mechanism. In terms of theory growth, Rosen's ideas were more foundational (what *must* any natural system have to be alive?) than generative (they don't describe how new functions evolve). So while **organizational emergence** is central (how a set of functions becomes self-sustaining), **open-ended evolution** per se was not explicitly modeled. The mapping to P1 (operator rewriting) is **speculative** here; one might consider the way metabolic reactions transform substrates as an "operator" algebra, but Rosen's math was category-theoretic, not a rewrite system. Overall, Rosen provides a conceptual framework where P5/P4 are explicit and necessary, and the other primitives can be read in implicitly. Emergence of a closed loop organization is a key inference (strong mapping), but using the framework predictively for evolving novelty remains an open challenge.

## Autocatalytic Sets and Chemical Organization – Kauffman (1970s) and Beyond

In the realm of origin-of-life studies and artificial chemistry, **autocatalytic set theory** offers a framework explicitly built on primitives like **closure**, **catalytic gating**, and **resource supply** – making it a strong match for P2, P5, P6, with implications for P3 and P4. Stuart Kauffman pioneered this idea: an *autocatalytic set* is a collection of molecular species and reactions such that the set **catalyzes its own production** (each reaction is catalyzed by some species in the set) and can sustain itself given a "food" supply of basic ingredients <sup>21</sup> <sup>22</sup>. This was later formalized as RAF theory (Reflexively Autocatalytic and Food-generated sets) by Hordijk, Steel, and others <sup>21</sup> <sup>22</sup>:

- **Catalytic Closure (P5/P2):** A RAF or autocatalytic set is *closed* under the reactions: all molecules produced are themselves members of the set (no outsiders needed) <sup>21</sup>. Moreover, it is *self-catalyzed*: *each reaction requires a catalyst that the set provides*. This embodies **closure (P5)** – the set, once established, produces no new chemical species beyond itself. And it embodies **constraint gating (P2)** very explicitly: *reactions only proceed if the appropriate catalyst is present*. Catalysts act like logical gates – without the catalyst, a reaction's pathway is practically "off." The presence of catalysts in the set turns reactions "on," enabling the network to replicate or sustain. Kauffman described this condition as **"catalytic closure"**, when a chemical network becomes collectively self-sustaining <sup>23</sup>. The RAF formalism precisely captures that notion: every reaction is catalyzed internally, and all required reactants can be obtained from within the set or a basic food source <sup>21</sup>.
- **Invariant Reaction Network (P4):** Once an autocatalytic set forms, it tends to exhibit *homeostatic stability*: the relative abundances of species might stabilize (a steady state) and the *network of reactions* acts as a unit. This network can be viewed as an **invariant class (P4)** – for example, it might correspond to a basic metabolic cycle that, as long as maintained, defines a "species" of protocell or chemical organization. Even if individual molecules turnover, the pattern of catalysis remains. In RAF theory, one can identify **strongly connected subsets** or hierarchical layers of autocatalytic sets (subRAFs) <sup>24</sup> <sup>25</sup>. These are akin to *conserved subsets* of reactions that might persist even as the overall system grows – reminiscent of invariants or sectors in a state space. Modern algorithms can find all such closed self-sustaining subsets in a reaction network <sup>26</sup> <sup>27</sup>, essentially mapping out invariant sectors (organizations) within the larger space of chemistry.

- **Protocol/Phase Dependence (P3):** While chemical reaction networks themselves are not sequential programs, the *formation* of an autocatalytic set can be history-dependent. It might require a certain sequence of random reactions or a particular order in which catalysts appear for the closure to “snap” into place. Once formed, the *dynamics* often entail **phases** like growth and division (if compartments are present) or cycles of production and depletion. For instance, an experimental autocatalytic set might go through an oscillatory phase (if some intermediate must accumulate before catalyzing another). **Order-dependence** also appears if multiple autocatalytic sets compete: the first set to establish can consume resources and prevent others (“locking in” one emergent order). This is analogous to **protocol dominance** – the early sequence of events biases the outcome (a symmetry-breaking event). Kauffman often emphasized that once a certain diversity threshold is passed, autocatalytic closure *inevitably* occurs, but *which* particular set forms can depend on chance seeding (historical contingency). In summary, P3 is present in a qualitative sense (path-dependent emergence of particular autocatalytic organizations).
- **Resource-Driven Asymmetry (P6):** Autocatalytic sets *require* a constant feed of basic “food” molecules (energy-rich substrates) from the environment <sup>28</sup> <sup>23</sup> . The RAF definition includes that all reactions ultimately trace back to a supplied food set <sup>28</sup> . This is clearly **resource-driven**: without continual supply, the reactions would equilibrate and stop. The need for external high-free-energy compounds means the system is **dissipative** and inherently time-asymmetric – it only persists by consuming available resources (like fuel) and producing waste. Kauffman and others note this as a principle: **open-ended evolution requires a non-equilibrium flow** to allow novelty (new reactions) to keep appearing <sup>18</sup> <sup>19</sup> . Indeed, experiments with autocatalytic networks (e.g. in RNA chemistry) show that feeding and dilution (to remove waste) are key to maintaining evolution rather than stagnation <sup>29</sup> .

Autocatalytic set theory thus has a **very strong mapping** to four of the six primitives (P2, P5, P6 explicit; P4 inherent; P3 qualitatively present). What about **operator rewriting (P1)**? In one sense, chemical reactions themselves are *rewriting rules*: reactants are transformed into products. Kauffman’s original models often used **rule-based combinatorics** (e.g. polymers concatenating and cleaving – effectively string rewrite rules) to simulate chemical novelty <sup>30</sup> . Furthermore, RAF theory has been generalized beyond chemistry: any process where “items” are transformed and catalysts facilitate transformations can be seen as a RAF-like system <sup>31</sup> . This includes abstract rewriting systems and even economic or cognitive processes <sup>31</sup> . Thus, one can interpret **P1 (operator rewriting)** in these models as the **generative production rules** (reaction formulas) that define what transformations are possible. The *discovery of new possible reactions* (e.g. new polymers forming new catalysts) is effectively the introduction of new rewrite rules into the system – enabling open-ended growth of complexity. While Kauffman didn’t frame it in formal language terms, the parallel is strong: chemical reaction networks and formal grammars both rely on rule-based generation. Indeed, Fontana and Buss (1994) recast chemistry in **lambda-calculus (a rewriting system)** to study emergent organizations <sup>32</sup> .

**Emergence:** Autocatalytic sets are explicitly a theory of **emergent self-organization** – e.g. the spontaneous appearance of a self-sustaining metabolism from a “soup” of molecules <sup>33</sup> . Kauffman famously argued this could explain the *origin of life* (an emergent autocatalytic metabolism) without external design. Because new species of molecules can form (especially in models allowing combinatorial polymer growth), this framework supports **open-ended evolution** to a degree – the space of possible molecules expands and can yield novel autocatalytic cycles. However, a known challenge is that a closed autocatalytic set can “take over” and suppress further novelty (by consuming resources). Researchers are actively exploring extensions like

**generative autocatalytic sets** (where smaller RAFs can combine into bigger ones) <sup>24</sup> <sup>25</sup> to allow sustained innovation. Nonetheless, **irreversibility** is built-in (due to resource flux and selection of one autocatalytic subset out of many possible), and **phase transitions** (P3) appear when the system crosses thresholds (sufficient diversity yields a phase change from no autocatalysis to autocatalysis). In summary, autocatalytic set theory/RAF is one of the clearest matches to the six primitives: **strong mapping** (explicit P2, P5, P6; well-covered P1, P4; moderate P3), and a direct focus on emergent, self-maintaining structures in open-ended chemical evolution <sup>34</sup> <sup>20</sup> .

## Chemical Organization Theory – Dittrich & Speroni (2000s)

Building on ideas from Fontana & Buss and autocatalytic sets, Dittrich and Speroni developed **Chemical Organization Theory** (COT) to formally identify emergent organizations in complex reaction networks <sup>27</sup> <sup>26</sup> . A *chemical organization* is defined as a set of species that is **closed** (no reactions produce species outside the set) and **self-maintaining** (each species in the set can be produced from others in the set) <sup>27</sup> . This concept refines P5 and P2 in a rigorous way:

- **Closure and Self-Maintenance (P5 & P2):** Closure here is precisely **P5 (closure)** – if you start with that set of molecules, no new molecule type will appear; it's closed under the chemistry <sup>27</sup> . Self-maintaining means for every species in the set, there is at least one reaction that produces it whose inputs are *also* in the set (or available buffer like food), ensuring the set can replenish itself <sup>35</sup> . Self-maintenance is effectively an **internal constraint** that gates viability: if a purported set had a species that nothing else produces (and it doesn't feed in externally), that species would decay – so it's not self-maintaining <sup>35</sup> . Thus only sets obeying certain *constraints* qualify. In this way, COT makes **constraint gating (P2)** an **explicit criterion** for an organization: all necessary production constraints must be satisfied internally <sup>35</sup> . Dittrich shows that from any large reaction network, one can algorithmically derive the *hierarchy of organizations* (closed, self-sustaining subsets) <sup>36</sup> . This hierarchy constitutes **invariant classes (P4)** – once the system's state falls into the basin of one organization, it will tend toward that subset's steady state. In fact, a theorem in COT proves that every fixed-point (stationary state) of the reaction dynamics corresponds to an organization <sup>37</sup> . Thus organizations are **attractors (invariant sectors)** in state space by design <sup>26</sup> <sup>38</sup> .
- **Dynamics and Phase Changes (P3):** COT explicitly connects network **dynamics to movement between organizations** <sup>26</sup> . As conditions change or as random fluctuations produce new species, the system's composition can leave one organization and enter another. These transitions can be seen as **phase changes** or regime shifts – e.g. introduction of a new species might create a larger closed set that “swallows” the current one. The sequence in which new species appear thus matters (path dependence, P3): if a critical catalyst arises too early it might not be sustained, but appearing after others are present, it may tip the system into a new organizational state. Dittrich's framework was not about temporal scheduling per se, but about **state-dependent regimes**. However, by mapping states to organizations, one implicitly gets a partial order (the “organization lattice”) which the system may climb as complexity grows <sup>36</sup> . The process of discovery of larger organizations as parameters change is analogous to **theory growth or open-ended evolution** in chemical space.
- **Resource Input (P6):** Like RAF, chemical organizations typically assume a set of basic food species that are externally buffered (maintained by environment). This is necessary for open systems. The presence of a food set means the **affinity** (driving force) comes from unlimited supply of some precursors. Organizations are defined with respect to that environment. If the environment changes

(different food or influx), the lattice of possible organizations can also change. So resource context explicitly drives which organizations can exist. The asymmetry of having an external “chemostat” (fixed concentration source) versus internal species that dilute if not produced makes the dynamics irreversible (the environment breaks detailed balance). Thus P6 is again inherently present.

The mapping of COT to P1 is subtle: it doesn't highlight “operator rewriting” except that the reaction rules themselves are the generative engine for state changes. However, since COT was *inspired by Fontana's lambda-calculus model* <sup>39</sup>, one could say it inherits an implicit *rewriting perspective* – molecules were like expressions that combine and reduce. In any case, COT's main contribution is formalizing P5, P2, P4 in service of understanding emergence of *new organizational structures*. It **strongly** covers those, and P6 by context. P3 is acknowledged in discussing how systems move *between* organizations (e.g. ecology succession from one stable community to another is one analogy).

**Emergence:** COT is aimed squarely at identifying emergent *meso-level structures* in complex chemical or biological systems <sup>27</sup> <sup>38</sup>. By finding organizational closures, it explains how new “wholes” can appear and persist. This has been applied to models of metabolism, immune dynamics, and even social simulations to detect self-organized groups <sup>38</sup>. Emergence here is **inferred via invariants**: an organization is an emergent set of entities with stable relationships. The framework implies that open-ended evolution would manifest as an ever-expanding chain of organizations (in a hierarchy or partially ordered set), where new larger closures form as smaller ones combine – an idea closely related to **meta-system transitions** (Turchin) or **increase in individuality** in evolution. The mapping of P1–P6 is **strong for P2, P4, P5, P6**, and **moderate for P3**. The **irreversibility** is captured by the idea that once a new higher organization forms and is stable, the system would need a significant perturbation to leave it – path-history matters. COT does not explicitly introduce a new rewriting rule mechanism (other than new species appearance), so P1 remains **weak** here.

## Synergetics & Dissipative Structures – Haken and Prigogine (1970s)

From theoretical physics, *Synergetics* (Hermann Haken) and the theory of *dissipative structures* (Ilya Prigogine) provide frameworks for emergence of order in far-from-equilibrium systems. They don't enumerate six primitives as a list, but key concepts in these theories align with several of P1–P6, especially **constraint (P2)**, **order parameters/invariants (P4)**, **sequence dependence (P3)**, and **resource gradients (P6)**:

- **Order Parameters & Enslaving (P4 & P2):** Haken's Synergetics introduced the idea of *order parameters* – collective variables that describe the emergent macroscopic pattern (e.g. the amplitude of a convection roll pattern in fluid) <sup>40</sup> <sup>41</sup>. These correspond to **invariant classes (P4)** in that once the system selects a particular pattern (an attractor), the order parameter's value characterizes the whole domain (e.g. all convection cells rotating clockwise vs counterclockwise is a distinct sector). Along with this, the “**enslaving principle**” says that the order parameters *enslave* the microscopic degrees of freedom: fast micro-variables adapt to the slow, dominant pattern <sup>41</sup>. This is effectively **constraint gating (P2)**: once an order parameter emerges, it **constrains the motion of components**, reducing freedom (e.g. molecules now circulate in an organized roll, not randomly). The allowed microstates are gated by the emergent structure (symmetry-breaking selects one mode and suppresses others). Haken showed how a few modes (order parameters) control the rest in



lasers, chemical oscillations, etc. Thus an emergent pattern imposes **internal constraints** that were not active before its appearance.

- **Bifurcation and Path Dependence (P3):** Both Haken and Prigogine studied **bifurcations** – critical points where a small change in a control parameter (e.g. temperature difference, or chemical flow rate) causes the system to switch from disorder to order or from one pattern to another <sup>42</sup> <sup>43</sup> . At these junctures, often the system can go into one of several possible *new states* (e.g. which direction will the convection cells rotate? Which chemical pattern forms?). The selection can be due to random fluctuations (“noise”) pushing it one way – hence **protocol or sequence matters**. If a fluctuation biases the choice, the system *locks into that pattern irreversibly* (hysteresis may prevent return to the old state without big changes). Prigogine called this **“order through fluctuations”** <sup>44</sup> – essentially the system’s history of fluctuations determines the emergent order, reflecting **path dependence (P3)**. Once a particular symmetry is broken, the system is in a new sector and can’t retrace without undoing the broken symmetry (which often requires a reverse sequence of changes that is unlikely). This is a direct analog of **phase-dependent outcomes** – e.g. *which attractor you end up in depends on the path taken*. Both theories emphasize the *irreversible selection* of a branch of solutions, embodying P3 strongly.
- **Invariants & Sectors (P4):** After a bifurcation, the system often settles into one of multiple **“attractor” states**, each with its own characteristics (e.g. pattern A vs pattern B). These can be seen as **distinct invariant sectors** of state space. Once in one, certain conservation-like rules apply. For instance, an oscillatory chemical pattern might have an invariant frequency. Or in a ferromagnet (another symmetry-breaking system), once magnetized “up” vs “down,” the system has a conserved bulk magnetization sign (an invariant property unless enough energy is input to flip it). Prigogine’s school even talked about **“structural stability”** – once formed, the structure resists small perturbations, maintaining its invariants. In some cases, near a bifurcation, **fluctuations themselves become scale-invariant** (critical phenomena) – which is another interesting connection to invariants (critical exponents, etc., though that’s more in equilibrium phase transitions). Overall, the emergent ordered states define **classes of organization** the system can have.
- **Resource-Driven Affinity (P6):** A *dissipative structure* (Prigogine’s term) is one that can only exist due to a continuous **throughput of energy or resources** <sup>45</sup> <sup>46</sup> . Classic examples: the Bénard convection rolls require heating from below (thermal gradient) <sup>47</sup> ; the Belousov-Zhabotinsky chemical oscillator requires an open flow reactor with reagents pumped in <sup>48</sup> . These gradients are sometimes called **thermodynamic affinities** – e.g. a chemical “force” driving reactions in one direction. If the flow/affinity drops below a threshold, the structure collapses back to disorder <sup>42</sup> . Thus **P6** is central: only by consuming free energy (creating entropy in the environment) can the system maintain lower internal entropy and organized patterns <sup>46</sup> . This one-way consumption imposes an **arrow of time** (irreversibility) on the process – the emergent pattern decays quickly if input stops. In Prigogine’s words, the system “exports entropy” to keep itself organized <sup>46</sup> .
- **Operator Rewriting (P1):** These physics frameworks don’t explicitly involve *operators rewriting each other*, as they are typically described by differential equations. However, one might draw an analogy: when the system goes through a bifurcation, the effective *dynamics* can often be rewritten in terms of new variables (order parameters) with simpler equations (“slaving” eliminates fast variables). This **change of variables** is like rewriting the description of the system in a new basis adapted to the emergent structure. It’s a stretch to call this operator rewriting, but it does resonate with the idea of

*reformulating the rules* at a new level (e.g. the post-bifurcation regime obeys a different effective law than the pre-bifurcation regime – perhaps analogous to how a rewrite rule might be replaced by another under certain conditions). This mapping is **speculative**. A more concrete computing analogy: one could simulate a pattern-formation system via cellular automata or lattice models where local *update rules* (operators) get effectively overridden or biased once a large-scale pattern appears (because the pattern creates long-range correlations). In that sense, the *local update rule outcomes are “rewritten” by global constraints*. Again, this is not standard phrasing in physics, but it connects to the spirit of P1.

In summary, Synergetics/dissipative structures provide a **strong conceptual match** to **P2 (constraints via order parameter enslaving)**, **P3 (path-dependent bifurcation selection)**, **P4 (distinct ordered phases as invariant regimes)**, and **P6 (need for a pump of energy/matter)**. They implicitly involve aspects of P1 (if one considers regime change as rewriting rules) and P5 (the emergent structure often forms a *closed pattern* like convection cells are spatially closed loops, or a limit cycle in chemistry is a closed trajectory in phase space). The mapping is thus **moderate to strong** overall.

**Emergence:** These frameworks were explicitly developed **to explain emergence of order from chaos**, and they succeed in describing how *irreversible processes lead to new ordered states* <sup>41</sup> <sup>46</sup>. Prigogine emphasized that **irreversibility is not just disorder; it can create new coherent structures** (contrary to the equilibrium thermodynamic intuition) <sup>41</sup> <sup>46</sup>. Both frameworks yield *qualitative novelty* (a new phase with new properties, e.g. “liquid becomes solid crystal” or “random fluid becomes patterned cells”). However, once an ordered state forms, the *set* of possible patterns is often fixed by symmetry – it’s emergent order, but not *open-endedly novel* beyond those bifurcations. To get open-ended evolution, one would have to have an endless cascade of bifurcations creating higher organization – which is not typical in simple physical systems (they often saturate at an ordered state given fixed conditions). Nonetheless, when coupled with other processes (e.g. ecological or evolutionary feedbacks that keep changing control parameters), these ideas can describe an ongoing emergence. In climate or ecology models, for example, multiple levels of pattern emerge as conditions evolve. In any case, the contribution of Haken and Prigogine was to identify core ingredients for emergence (nonlinearity, feedback, symmetry-breaking, energy flow) that closely parallel P1–P6. The **arrow of time (irreversibility)** is especially highlighted by the need for P6 resource flux and P3 historical selection. This is a **strong mapping in conceptual terms** and foundational for later interdisciplinary work on emergence.

## Open-Ended Evolution Constraints – Solé *et al.* (2024)

A very recent framework by Solé, Kempes, and colleagues attempts to outline fundamental **constraints on living systems** that shape what forms can emerge <sup>49</sup> <sup>50</sup>. While not framed as “primitives” of a generative system, their six highlighted aspects of life strongly echo our six primitives, and they explicitly connect to *open-ended evolution, convergence, and path-dependence*. The six aspects they discuss are <sup>49</sup>:

1. **Thermodynamic properties of living systems** – Life operates as a thermodynamic engine, far from equilibrium. (This maps to **resource-driven asymmetry (P6)**: the need for continual energy flow and entropy export, as well as the irreversibility and directionality that imposes <sup>51</sup>.)
2. **The linear nature of molecular information** – Genetic information is stored in linear polymers (DNA/RNA), implying sequential, one-dimensional encoding of biological complexity. (This relates to **protocol/order-dependence (P3)** and **operator rewriting (P1)**: the genetic code is read and

rewritten sequentially, and the linear arrangement means that *order of symbols matters* for function. It also imposes a unidirectional reading frame – an inherent protocol for gene expression.)

3. **The cellular nature of life's building blocks** – Life is made of discrete cells, each an enclosed compartment. (This is clearly **closure or packaging (P5)**: the cell membrane is a boundary that encapsulates an autonomous unit <sup>51</sup>. It's also **constraint gating (P2)**: the membrane gates what goes in/out, and cells partition reactions into separate spaces, limiting interactions.)
4. **Multicellularity and development** – Many organisms consist of multiple cells and go through developmental programs. (This involves **protocol/phase dependence (P3)**: development follows an ordered sequence of gene expression and differentiation events – a temporal program that must be executed correctly for a complex body to emerge. It also introduces hierarchical closure (cells into tissues, etc., another **P5**), and new constraints (**P2**: e.g. cells signaling to each other constrain fates). The presence of a developmental *program* explicitly highlights sequence-dependent emergence of form, which is a kind of controlled irreversibility.)
5. **Threshold nature of computations in cognitive systems** – Neurons and neural circuits often behave in an all-or-none, thresholded manner. (This maps to **constraint gating (P2)**: a neuron fires only if input sum exceeds a threshold, acting like a gate. Populations of neurons show phase transitions between activity states, and signals propagate in a protocol-like way through neural circuits (a timing-dependent process – hint of P3). It also underscores how **nonlinear switching** underlies brain emergent properties, which resonates with P2 gating and P3 sequence sensitivity – e.g. the order of neuron firing can encode information.)
6. **Discrete nature of ecosystem architecture** – Ecosystems show discrete units (species, trophic levels, functional niches) rather than continuous blends. (This corresponds to **invariant classes or sectors (P4)**: species are an invariant category – individuals come and go, but the species (as a gene pool) persists. Trophic levels (producer, consumer, etc.) are discrete functional classes in ecosystems, which remain identifiable even as specific players change. Ecosystem assembly follows certain rules (e.g. energy pyramids) that act like constraints on which configurations are viable – linking to P2 and P6 because resource flows between levels are conserved. Also, ecosystems can exhibit multiple stable states – distinct community structures – which are like alternative invariant sectors selected by history (P4 with P3).)

Solé *et al.* argue that these fundamental features both **limit and enable evolution** – they are why life on Earth (and perhaps anywhere) converges to certain forms (cells, genomes, networks) despite historical contingency <sup>52</sup> <sup>53</sup>. In terms of our six primitives, this list is essentially a direct mapping: - P6: thermodynamic engine (non-equilibrium drive). - P3: path-dependent, sequential information (linearity of DNA, developmental sequences). - P5: compartmentalization (cells). - P2: thresholds (gating in neural and other systems). - P4: discrete hierarchical organization (cells, species, ecosystem modules). - P1: one might see the “computations” and information encoding as a form of processing operations (e.g. DNA transcription/translation as rule-based decoding, neural computations as transforming signals) – though Solé *et al.* don't explicitly mention rewriting, the idea of *digital (discrete) information* suggests that logical operations (like the genetic code's codon->amino acid mapping) act as **operators** in biology. One could also frame developmental rules or ecosystem assembly rules as “operators” that rewrite an embryonic state into a mature form, or an empty habitat into a climax community.

The authors explicitly highlight **emergence and path-dependence**: “the historical nature of evolution makes it highly path-dependent... yet convergence and constraints limit the domain of possible designs” <sup>52</sup>. They propose that these six factors are “fundamental constraints... associated with the logic of living matter” <sup>54</sup>. In other words, while evolution is open-ended, it is *channeled* by these recurring primitives. This framework is less a generative *model* than an analytic one, but it effectively identifies which primitives must be in any generative theory of open-ended evolution.

**Emergence:** Very high emphasis. Each item is discussed in terms of how it allows complex emergence: e.g. thermodynamics + self-organization, developmental programs enabling multicellular emergence, neural gating enabling cognitive emergence, etc. <sup>49</sup> <sup>50</sup>. They also clearly note **irreversibility and contingency** (path-dependence) <sup>52</sup> alongside **convergence** (certain patterns keep re-emerging because of the constraints). The mapping to P1–P6 is **strong** (since they practically listed six analogous concepts), though P1 is the least explicit in their text (it lurks behind the scenes as the digital/program-like aspect of biology). This work is an *integrative interdisciplinary perspective*, covering physics (thermodynamics), biology (molecular to ecosystem), and cognitive systems, exactly as the question prompt anticipated. It suggests that any comprehensive theory of open-ended evolution and emergence will involve something like **P1–P6 as foundational “ingredients.”**

## Other Analogous Frameworks and Mappings

In addition to the major frameworks above, numerous concepts across fields implicitly group similar primitives as generative mechanisms. We briefly highlight a few, noting the strength of the analogy:

- **Generative Grammars (Chomsky, 1950s–)** – In linguistics, generative grammar uses **production rules** (rewrite rules, P1) to generate sentences, and applies **constraints** (like syntactic or semantic filters, P2) to rule out ill-formed ones. The derivation of a sentence is **order-dependent (P3)** – the sequence of rule applications (and transformations in earlier models) matters. Grammars define **invariant classes (P4)** such as syntactic categories (Noun, Verb, etc.) that remain constant throughout derivations, and use **hierarchical closure (P5)** by embedding phrases within phrases (packaging sub-structures). In later theories (Principles & Parameters, Minimalist Program), resource-sensitive conditions like the *Shortest Move* or *derivational economy* act like **affinities (P6)** favoring certain derivations. While linguistics isn’t about physical emergence, it *is* about how infinite novel structures (sentences) emerge from finite rules – an **analogous generative emergence**. The mapping P1–P5 is strong in classical generative grammar, and P6 is weaker (it appears only metaphorically, e.g. in considering processing limitations). This is a **speculative mapping** to real emergence: one might say the evolution of language or the growth of a child’s grammar is an emergent, open-ended process modeled by such rules, but grammar frameworks themselves are largely static competencies.
- **Constraint Logic Programming & AI Planning (1980s–)** – In computer science, constraint logic programming (CLP) and related paradigms combine **rules** (logic clauses, which we can see as rewrite/derivation rules, P1) with **constraints** (P2) that prune the search space of solutions. The solver’s **search strategy** introduces a kind of **protocol dependence (P3)** – the order in which subgoals or variables are addressed can affect performance or even outcome (for non-confluent search). Solutions that satisfy all constraints can be seen as **invariant end-states (P4)** of the program (e.g. a solved puzzle is an invariant configuration that all execution paths must reach). Many CLP systems encapsulate subproblems or use modules, providing **packaging (P5)**. Some CLP

and AI systems incorporate **resource limits or costs**, effectively adding a **preference or bias (P6)** guiding which solutions are found first (e.g. a cost function that acts like an affinity for cheaper solutions). While these frameworks are **algorithmic rather than descriptive of nature**, they have been used to model planning and adaptive processes. The mapping of P1–P6 is **moderate** here, but **emergence is limited**: typically, CLP finds a predefined goal state rather than generating unbounded novelty. It's more an analogy of mechanism (rules + constraints + search) to natural evolution than a direct emergence framework.

- **Category Theory and Monads (1960s–)** – In category theory applied to computer science (and even in some systems science metaphors), a **Monad** is an abstraction that **encapsulates computations with context**. Monads have an operator (bind) that sequences operations (**enforcing order, P3**), threading a hidden context (state, effects, etc.) through them (**packaging context, P5**). The monad laws impose **invariants (P4)** on how the operations compose, and often monads include unit and binding conditions that **gate what computations are valid (P2)**. For instance, a “Maybe” monad stops the computation if an error occurs – gating further operations based on a constraint (presence of a value). Monads can also model resource consumption (e.g. an I/O or state monad carries a world state that gets changed – an implicit **resource-affinity, P6** toward consuming the world state as you go). While monads are a purely formal construct for structuring programs, the *idea* of a standardized **closure operation (return) and sequence operator** has been compared to processes in nature that “package” intermediate results and pass them along. This analogy is **weak** but conceptually intriguing: one could imagine an emergent process as a composition of monad-like transformations, where each step's output is fed as input to the next under certain constraints. Some cybernetic models have even used category theory to describe living systems' organizations (e.g. **Memory Evolutive Systems** by Ehresmann & Vanbreemersch, which employs category theory to formalize hierarchical closures and regulatory gates). In such models, the mapping to P1–P6 can be strong, but it becomes fairly abstract. **Emergence** in category-theoretic terms usually appears as *higher-level functors or limit objects* that encapsulate lower-level interactions – again reflecting closure (colimits), constraints (pullbacks, equalizers), etc., but this remains in the realm of mathematical analogy.

- **Spencer-Brown's “Laws of Form” (1969)** – This is a self-contained calculus of distinctions. It starts from a single primitive operator (the “mark”, drawing a distinction) and generates forms through iterative application (which is literally **operator rewriting (P1)** on a diagrammatic/algebraic level). A marked distinction inherently creates an inside and outside, introducing an **asymmetry (P6)** – the marked side vs unmarked. The calculus has rules that maintain certain **invariants (P4)** (e.g. “law of calling” and “law of crossing” yield invariant equivalence classes of forms). One could interpret the successive drawing of distinctions as a **sequence-dependent protocol (P3)** for constructing complex forms, and each closed contour as an **encapsulation (P5)**. Some thinkers (e.g. G. Spencer-Brown, Louis Kauffman) have speculated that this formalism underlies cognitive or cosmic emergence – essentially that *making distinctions* is the fundamental act of creation of new structure. However, this remains **highly speculative**; it's a philosophical analogy more than an empirically grounded framework. The “laws of form” certainly embody P1 and P5 (and arguably P6 in the initial act of distinguishing something from void), but whether they explain open-ended evolution or irreversibility is dubious. It might be better to see it as a *metaphor* for how from a void (symmetric state) the first distinction (symmetry break) creates the first bit of information, and subsequent distinctions build hierarchies – a creation myth in algebraic clothing. The mapping strength is **weak**, and emergence is only inferred philosophically (the calculus itself is reversible algebraically, though

the interpretation of a distinguished mark vs blank space is an irreversibly broken symmetry of the paper).

- **Evolutionary Epistemology & Theory Change (1960s–1970s)** – In the philosophy of science, ideas from Popper, Campbell, and later Dennett and others have likened the development of knowledge and scientific theories to evolutionary processes. Donald Campbell's *Blind Variation and Selective Retention* (BVSr) model posits that new ideas are generated by somewhat random variation (**operator variation akin to rewriting hypotheses, P1**) and then filtered by empirical tests or logical consistency (**selection constraints, P2**), and the surviving ideas are retained (forming an **invariant core of knowledge, P4** until the next change). Over time, scientific fields show **order-dependent development (P3)** – e.g. you must discover certain foundational concepts before others; also path-dependence in technology (QWERTY effect, etc.). Kuhn's notion of *paradigm shifts* can be seen as **phase transitions** where the “normal science” within a paradigm is an invariant sector, and a crisis triggers a reorganization to a new paradigm (a different invariant set of assumptions). Each paradigm has its own **closure (P5)** – questions it considers legitimate, tools it uses – and they are **incommensurable** to some degree (reflecting encapsulation). The process is **resource-driven (P6)** in a metaphorical sense: for example, funding or social needs drive certain research directions (introducing asymmetries in what gets pursued). While these ideas are not formalized with the six primitives explicitly, one can map the analogies fairly well. **Emergence of theories** and open-ended growth of knowledge are explicitly the focus. The mapping strength is **moderate analogical**. For instance, Campbell's evolutionary epistemology clearly identifies variation (new conjectures) and selection (experiments) as drivers – matching P1 and P2 – and implies that over time we accumulate a corpus of knowledge (like a well-adapted schema, P4) and occasionally undergo conceptual revolutions (path-dependent phase shifts, P3) that often require new conceptual “vessels” (new jargon, methods – akin to P5 encapsulating the new theory). These frameworks reinforce that *any evolving system that produces novelty while preserving gains* will likely involve generative rules, constraints, cumulative stable units, cycles of accumulation and revolution – much like P1–P6.

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**In conclusion**, a survey across disciplines finds **convergent support for the six primitives** as key ingredients in frameworks of emergence and open-ended evolution. Complexity science (CAS) and artificial life explicitly embrace rule-based generation (P1), selective constraints (P2), historical/phase-dependent dynamics (P3), emergent invariants (attractors, P4), modular or recursive closure (P5), and thermodynamic or resource-driven arrows (P6) in explaining how novelty and self-organization occur <sup>1</sup> <sup>21</sup>. Systems biology and origin-of-life theories (autopoiesis, autocatalytic sets, organization theory) explicitly build on closure and catalytic gating to yield self-sustaining networks <sup>16</sup> <sup>30</sup>, requiring energy throughput to persist <sup>46</sup>. Non-equilibrium physics formalizes how energy flow and nonlinear feedback break symmetries (constraints) to produce new ordered phases with distinct invariants <sup>41</sup> <sup>46</sup>. Even cognitive and social systems theories implicitly rely on similar mechanisms (thresholds, communication protocols, self-referential closure) when tackling emergent phenomena.

Where the mapping is **strong**, the frameworks themselves list or incorporate these principles (as we saw with autopoiesis, RAF, Solé et al.'s constraints <sup>51</sup>). Where it is **weaker**, we still find that interpreting the framework through P1–P6 sheds light on commonalities. Notably, **emergence** is a unifying theme: each framework, in its domain, tries to explain how complex, larger-scale or novel structures arise that are *more than the sum of their parts* – whether that be a coherent ant colony, a living cell, a new chemical metabolism,

a convective roll, a mind's thought, or a scientific paradigm. They achieve this by invoking *generative operations plus constraints, phases and closures, conservation of some aspects and consumption of others*.

Finally, the **strength of each mapping** varies: frameworks like RAF/COT and autopoiesis directly realize most of P1–P6 (strong mapping), whereas analogies like grammar or monads capture the form of the logic but not the creative openness (weak mapping). Speculative connections (Laws of Form, etc.) hint that even at abstract levels, one sees echoes of “distinctions and closures” making reality from blankness, but these remain hypothetical in explaining real-world open-ended evolution.

In summary, **many modern frameworks (post-1950s)** across disciplines either enumerate or implicitly use a set of fundamental operations and controls that correspond closely to the six primitives. These primitives in concert enable systems to generate novelty *and* channel it into higher-order stable structures – the essence of emergence and evolving complexity. The recurrence of these concepts – from enzymes catalyzing metabolic loops <sup>23</sup>, to tags enabling agent cooperation <sup>4</sup>, to cells compartmentalizing genetic circuits <sup>51</sup>, to energy gradients driving new order <sup>46</sup> – suggests that the mapping is **robust**: it is not an accident but rather reflective of deep principles governing how *any* complex adaptive system must operate to continually create and organize complexity. Each framework provides its own terminology and emphasis, but the overlap with P1–P6 is unmistakable and often explicitly noted in the literature (as seen in the cited sources). Thus, the six primitives appear to form a broadly applicable lexicon for understanding generative mechanisms behind emergent, irreversible, and open-ended processes.

#### Sources:

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- Haken, H. – *Synergetics*: order parameters enslaving microdynamics (constraints from emergent order) <sup>41</sup>
- Solé, R., et al. (2024) – *Constraints on the Logic of Life*: identifies thermodynamics, linear genetic info, cellular units, developmental programs, neural thresholds, and ecosystem discreteness as fundamental factors shaping evolution <sup>49</sup> <sup>50</sup>. These map directly to P1–P6 as discussed.

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<sup>1</sup> <sup>2</sup> <sup>3</sup> <sup>4</sup> <sup>5</sup> <sup>6</sup> <sup>7</sup> <sup>8</sup> <sup>9</sup> <sup>10</sup> <sup>11</sup> <sup>12</sup> <sup>13</sup> <sup>14</sup> Hidden Order: How Adaptation Builds Complexity by John Holland – The Rabbit Hole

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